

Map in C++ Standard Template Library (STL)

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What are Maps in C++

Maps in C++ are associative containers that store elements formed by a combination of a *key value* and a *mapped value*.

Consider the example:

Key	Value
U19002	Ram
U19003	Aziz
U19005	Krishna
U19006	David

The above example shows a key and value pair. The Student ID is the key and each student has a different ID, hence unique key representing them.

Syntax:

```
map<key_type , value_type> map_name;
```

1. This is a basic syntax for creating a map in C++.
2. We have a key value of type `key_type` and a value associated with the key of the type `value_type`.
3. When we enter the values, they should be entered in a pair and we cannot enter them one by one.

Basic Functions Associated with Map

[begin\(\)](#) – Returns an iterator to the first element in the map

[end\(\)](#) – Returns an iterator to the theoretical element that follows last element in the map

[size\(\)](#) – Returns the number of elements in the map

[max_size\(\)](#) – Returns the maximum number of elements that the map can hold

[empty\(\)](#) – Returns whether the map is empty

[pair insert\(keyvalue, mapvalue\)](#) – Adds a new element to the map

[erase\(iterator position\)](#) – Removes the element at the position pointed by the iterator

[clear\(\)](#) – Removes all the elements from the map